WAP to Generate Fractal Images Using Computer Graphics.

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>

/* Function to draw Sierpinski Triangle */

void fractal(int x1, int y1, int x2, int y2,
          int x3, int y3, int n)
{
   int mx1, my1, mx2, my2, mx3, my3;

   if(n == 0)
   {
      line(x1, y1, x2, y2);
      line(x2, y2, x3, y3);
      line(x3, y3, x1, y1);
      return;
   }

   /* Midpoints */

   mx1 = (x1 + x2) / 2;
   my1 = (y1 + y2) / 2;

   mx2 = (x2 + x3) / 2;
   my2 = (y2 + y3) / 2;

   mx3 = (x3 + x1) / 2;
   my3 = (y3 + y1) / 2;

   /* Recursive calls */

   fractal(x1, y1, mx1, my1,
         mx3, my3, n-1);

   fractal(mx1, my1, x2, y2,
         mx2, my2, n-1);

   fractal(mx3, my3, mx2, my2,
         x3, y3, n-1);
}

void main()
{
   int gd = DETECT, gm;
   int n;
```

```
    clrscr();

    printf("Enter recursion level (1-5): ");
    scanf("%d", &n);

    if(n < 1 || n > 5)
    {
       printf("Invalid level");
       getch();
       return;
    }

    initgraph(&gd, &gm, "C:/TURBO C3/BGI");

    setcolor(WHITE);

    fractal(320, 50, 100, 400,
          540, 400, n);

    getch();
    closegraph();
}
```